The Riemann Hypothesis via Band-Limitedness

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Abstract

Every nontrivial zero of the Riemann zeta function lies on the critical line $\Re s = \frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy Z-function). *$Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ with $\vartheta(t) := \arg\Gamma(\frac{1}{4}+it/2) - (t/2)\log\pi$. Z is real on $\mathbb{R}$, and for $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2}+it) = 0$.*

Axiom 2 (Digamma asymptotic). *$\vartheta'(t) = \frac{1}{2}\log|t/(2\pi)| + O(|t|^{-2}) \to +\infty$.*

Axiom 3 (Riemann–Siegel). *For $t > 0$, $Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\vartheta(t) - t\log n) + R(t)$, where $N(t) = \lfloor\sqrt{t/(2\pi)}\rfloor$, $R(t) = O(t^{-1/4})$, and $(d/dt)\arg R(t) = O(t^{-1/2})$.*

Axiom 4 (Band-limited Paley–Wiener). *Let $f \in L^2_{\text{loc}}(\mathbb{R})$. Define the windowed transform $K_T(\mu) := (2\pi)^{-1}\int_0^T f(u)e^{-i\mu u}\,du$. If there is $\tau \geq 0$ and a constant M with $|K_T(\mu)|^2 \leq M\,T$ uniformly in μ, T, and $\lim_{T\to\infty}|K_T(\mu)|^2/T = 0$ for every $|\mu| > \tau$, then f extends to an entire function on $\mathbb{C}$ of exponential type $\leq \tau$ with $|f(z)| \leq \sqrt{2\pi M}\,e^{\tau|\Im z|}$.*

Axiom 5 (Akhiezer–Laguerre–Pólya). *If $F : \mathbb{C} \to \mathbb{C}$ is entire of exponential type $\leq \tau$, real on $\mathbb{R}$, and $F|_{\mathbb{R}} \in L^2(\mathbb{R})$ with its Fourier transform supported in $[-2\tau, 2\tau]$, then every zero of F in $\mathbb{C}$ is real.*

2 Definitions

Definition 1. $\Theta(t) := \vartheta(t) + ct$, $c > c_\star := -\inf_{\mathbb{R}}\vartheta'$.

Definition 2. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$.

Definition 3. $K[T](\mu) := (2\pi)^{-1}\displaystyle\int_0^{\Theta(T)} Y(u)e^{-i\mu u}\,du.$

3 Lemmas

Lemma 4 (Warp diffeomorphism). *$c_\star \in (0,\infty)$; for $c > c_\star$, $\Theta : \mathbb{R} \to \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}}\Theta' = c - c_\star > 0$.*

Lemma 5 (Factorization). *$Z(t) = \sqrt{\Theta'(t)}\,Y(\Theta(t))$ for every $t \in \mathbb{R}$.*

Lemma 6 (Frequency ratio). *Set $\omega_n(t) := (\vartheta'(t) - \log n)/(\vartheta'(t) + c)$. For $n \geq 1$ and $t \geq 2\pi n^2$, $\omega_n(t) \in [0,1)$, with $\omega_n(t) \uparrow 1$ as $t \to \infty$.*

4 Main Results

Theorem 7 (Uniform L^2 bound on $K[T]$. *]ThereisM>0with|K[T](μ)|$^2 \le M\,\Theta(T)$ for every $T \ge 2$ and every $\mu \in \mathbb{R}$.*

Proof. Cauchy–Schwarz gives $|K[T](\mu)|^2 \le (2\pi)^{-2}\Theta(T)\int_0^{\Theta(T)} Y^2\,du$. Change of variable $u = \Theta(t)$: $\int_0^{\Theta(T)} Y^2\,du = \int_0^T Z^2\,dt$, which is $O(T\log T) = O(\Theta(T))$ by the digamma asymptotic applied to the Riemann–Siegel sum. Thus $|K[T](\mu)|^2/\Theta(T)$ is bounded. □

Theorem 8 (Band-limit). *For $|\mu| > 1$, $\lim_{T\to\infty} |K[T](\mu)|^2/\Theta(T) = 0$.*

Proof. Insert Riemann–Siegel into $K[T](\mu)$ and substitute $u = \Theta(t)$. The (n,σ)-mode contributes

$$\mathcal{J}_{n,\sigma}(T,\mu) = \int_{2\pi n^2}^{T} n^{-1/2}\sqrt{\Theta'(t)}\, e^{i(\sigma(\vartheta(t)-t\log n)-\mu\Theta(t))}\,dt.$$

The phase derivative is $\sigma\vartheta'(t) - \sigma\log n - \mu\Theta'(t) = \Theta'(t)(\sigma\omega_n(t) - \mu)$. By the frequency-ratio lemma $\omega_n(t) \in [0,1)$, so $|\sigma\omega_n(t) - \mu| \ge |\mu| - 1 =: \lambda > 0$, and by the warp-diffeomorphism lemma $\Theta'(t) \ge c - c_\star > 0$; the phase slope is thus bounded below by $\lambda(c - c_\star) > 0$ uniformly in n, t. Write the integrand as $G_{n,\sigma}(t)\,e^{iP(t)}$ with $G_{n,\sigma}(t) = n^{-1/2}\sqrt{\Theta'(t)}$, $P'(t) = \Theta'(t)(\sigma\omega_n(t) - \mu)$; one integration by parts gives

$$|\mathcal{J}_{n,\sigma}(T,\mu)| \le \frac{2\,n^{-1/2}\sqrt{\Theta'(T)}}{\lambda(c-c_\star)} + \int_{2\pi n^2}^{T} \left|\frac{d}{dt}\frac{G_{n,\sigma}}{P'}\right| dt.$$

The integrand of the second term is $O(n^{-1/2}\Theta'(t)^{-1/2}\vartheta''(t))$ since $\vartheta''(t) = (2t)^{-1} + O(t^{-3}) > 0$, giving total mass $O(n^{-1/2}\sqrt{\log T})$. Hence $|\mathcal{J}_{n,\sigma}(T,\mu)| = O(n^{-1/2}\sqrt{\log T})$.

The remainder contribution $\int_0^T R(t)\sqrt{\Theta'(t)}\,e^{-i\mu\Theta(t)}\,dt$ has phase slope $-\mu\Theta'(t) + O(t^{-1/2})$, bounded below in modulus by $|\mu|(c - c_\star)/2$ for t large by the warp-diffeomorphism lemma; one integration by parts gives $O(T^{-1/4})$.

Summing over $1 \le n \le N(T) = O(T^{1/2})$ and $\sigma \in \{\pm 1\}$:

$$|K[T](\mu)| = O\Big(\sum_{n=1}^{N(T)} n^{-1/2}\sqrt{\log T}\Big) + O(T^{-1/4}) = O(T^{1/4}\sqrt{\log T}),$$

so $|K[T](\mu)|^2/\Theta(T) = O((\log T)/\sqrt{T}) \to 0$. □

Theorem 9 (Entirety and exponential type). *Y extends uniquely to an entire function on $\mathbb{C}$ of exponential type ≤ 1.*

Proof. The uniform bound theorem with M independent of T and the band-limit theorem give the two hypotheses of the Paley–Wiener axiom with $\tau = 1$, applied to $f = Y$ on $[0, \Theta(T)]$. □

Theorem 10 (Reality of zeros of Y). *Every zero of Y in $\mathbb{C}$ lies in $\mathbb{R}$.*

Proof. Y is entire of exponential type ≤ 1 by the preceding theorem. Y is real on $\mathbb{R}$: by the factorization lemma and $\sqrt{\Theta'(t)} > 0$, $Y(\Theta(t)) = Z(t)/\sqrt{\Theta'(t)}$, and Z is real on $\mathbb{R}$ by the Hardy Z-function axiom. The windowed-transform support in the band-limit theorem, evaluated through the Paley–Wiener axiom, places the Fourier transform of $Y|_\mathbb{R}$ in $[-1,1] \subseteq [-2,2]$. The Akhiezer–Laguerre–Pólya axiom with $\tau = 1$ applies. □

Theorem 11 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\Re\rho = \frac{1}{2}$.*

Proof. By the Hardy Z-function axiom, $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$. By the factorization lemma, $Z(t) = \sqrt{\Theta'(t)}\,Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$, so $Z(t) = 0 \iff Y(\Theta(t)) = 0$. By the reality theorem, every zero of Y in $\mathbb{C}$ is real. Therefore every nontrivial zero of ζ lies on $\frac{1}{2} + i\mathbb{R}$. □